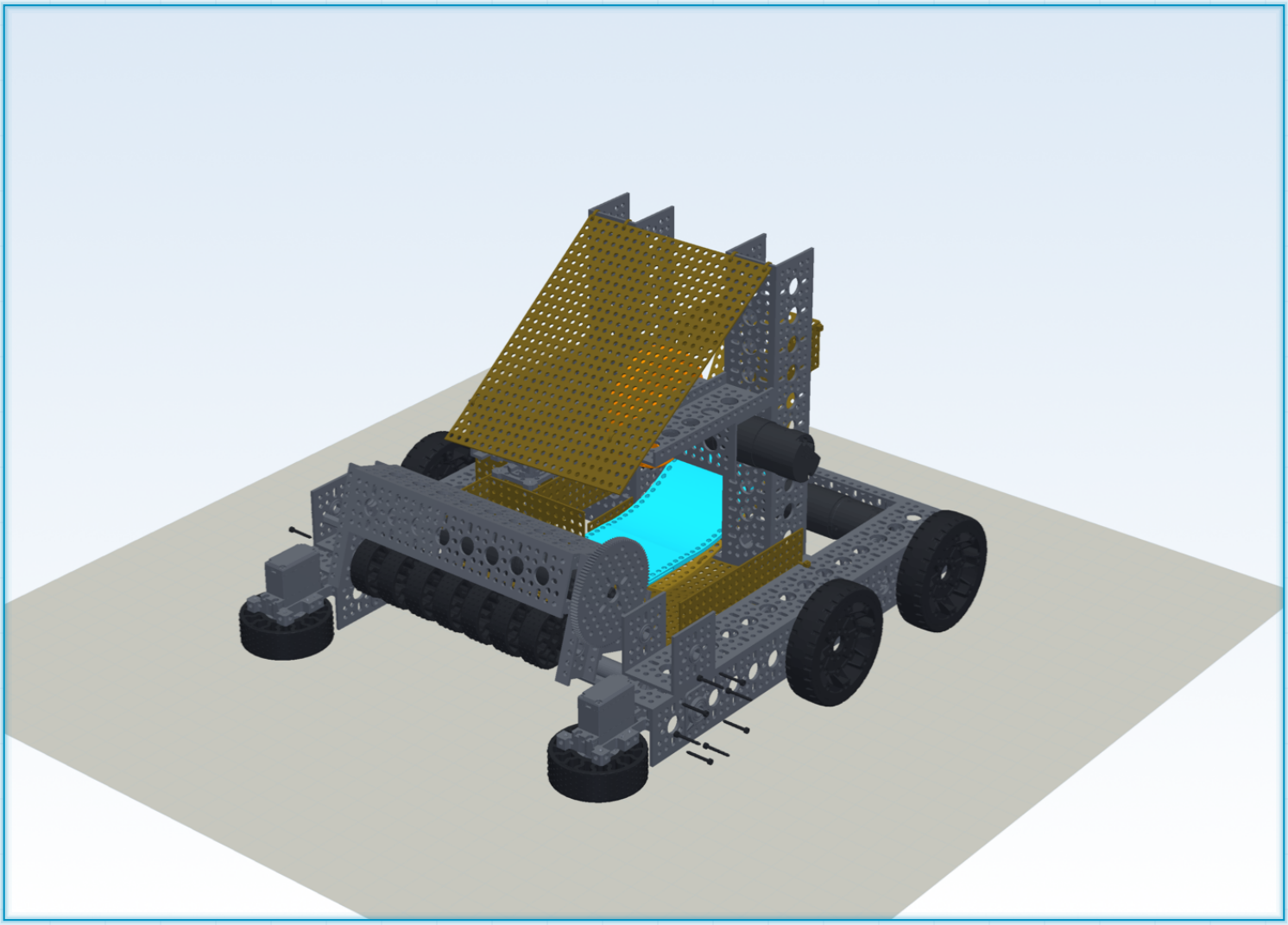
RAIL // GUARD

120 mm-radius shooting rail liner for the goBILDA FTC StarterBot (2026-27)

Stock 1:1 Yellow Jacket + 96 mm Hogback on Hyper Hub stay exactly as built. Zip-tie on, adjustable.



MISSION

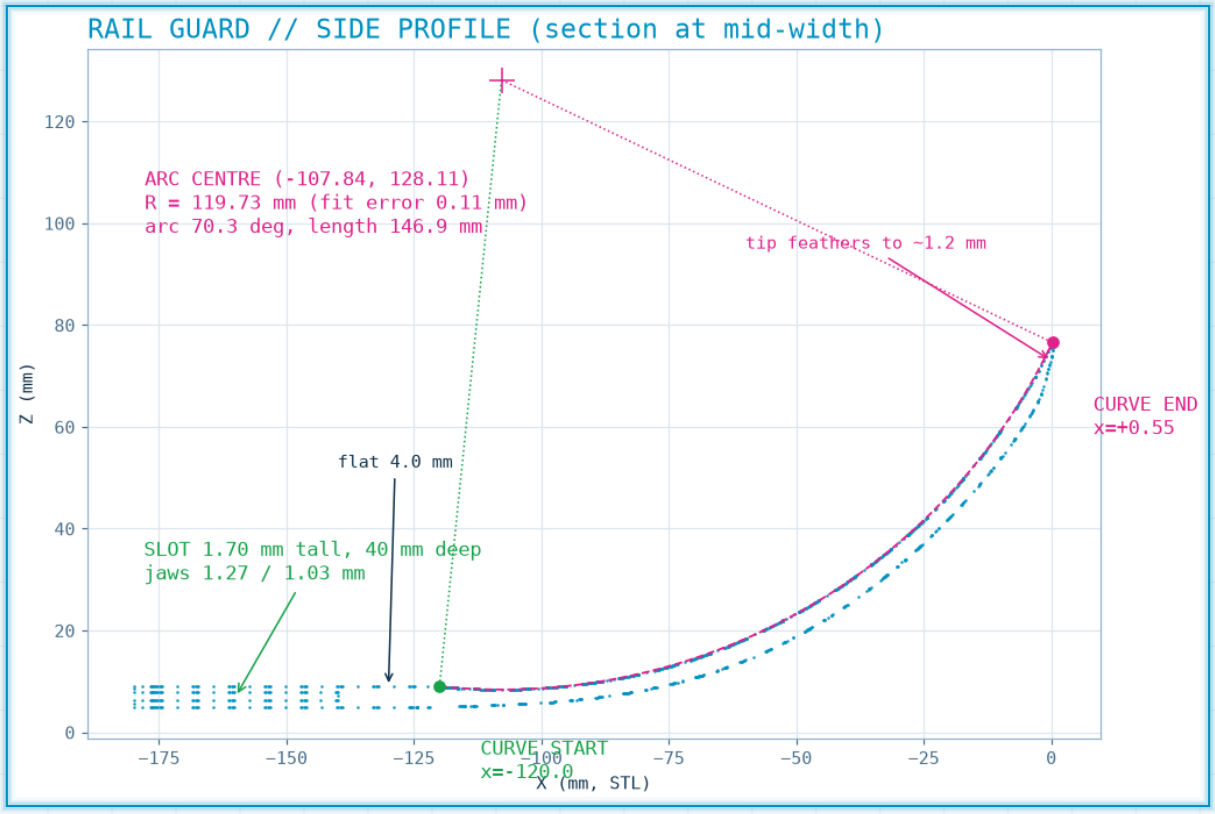
Flat section LEVEL with the ground; POLLEN rides the true 120 mm arc with 11-14 % squeeze.

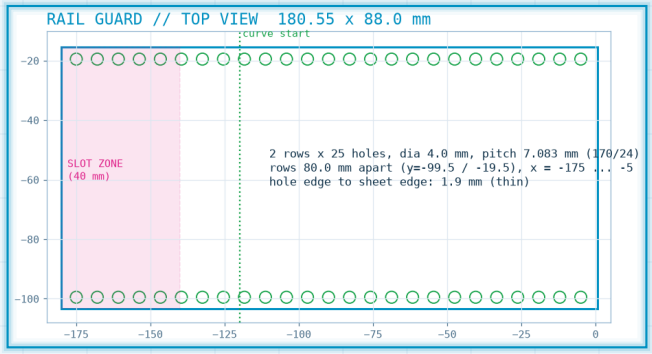
FILE

rail_guard.stl 180.55 x 88.0 x 71.3 mm
watertight, 1 body, no changes made

PART DATA

Measured from rail_guard.stl (unchanged)

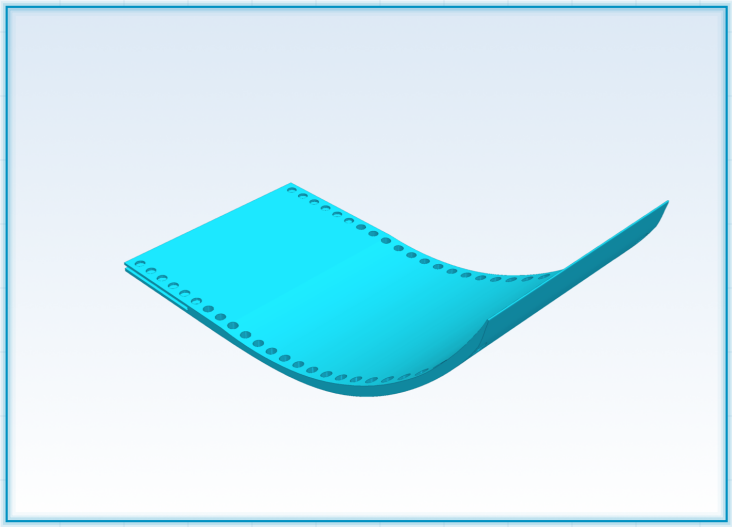


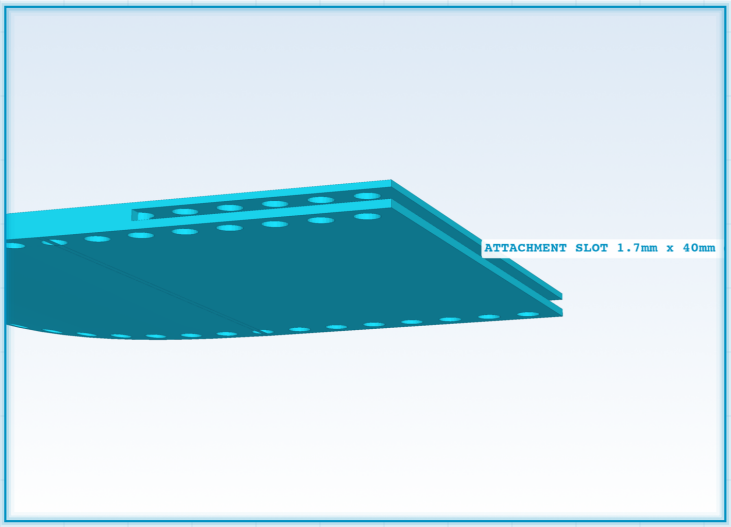


ARC: inner (ball) face is a true circle, R = 119.73 mm (best fit, max error 0.11 mm) from X = -120.0 to the tip at X = +0.55: 70.3 deg, 146.9 mm long. Bottom face is not concentric, so wall thickness varies 2.9-3.9 mm along the arc.
CURVE START/END: no cut or raised marking lines were found in the mesh. The start shows only as a faint 5.7 deg crease across the top at X = -120 (the flat meets the arc there; the arc's lowest point is 0.6 mm lower, 12 mm past the crease). The end is the tip itself. Lines at X ~ -125 on the underside are just mesh face splits and will not print.
ATTACHMENT: 1.70 mm slot, 40 mm deep, open at the front end (jaws 1.27 / 1.03 mm) - takes a ~1.5 mm plate edge. Superglue optional.
ZIP-TIE HOLES: 2 rows x 25, dia 4.0 mm, pitch 7.083 mm, rows 80 mm apart; only 1.9 mm of plastic between hole and edge.

PRINT SETUP

Best orientation: on its side (flat side face on the bed)

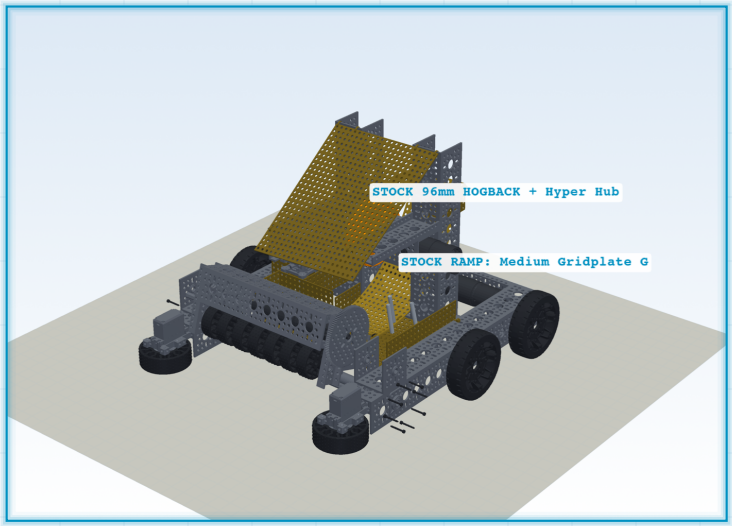




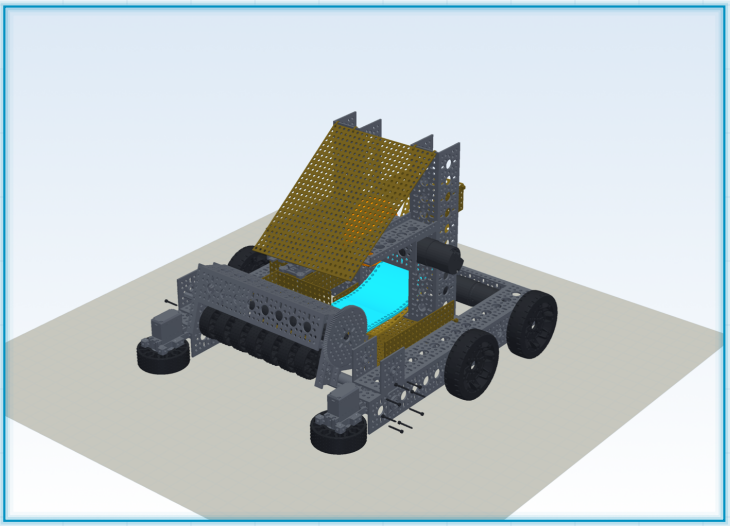
- Print STANDING ON ITS SIDE (the 180 x 71 mm side face down). The whole arc then prints with no overhangs, the slot is a simple vertical gap, and layer lines run along the ball's direction of travel (smooth).
- Lying flat it needs support under the curve and inside the 1.7 mm slot, which is very hard to remove.
- Use a brim (narrow footprint). 4 walls, 30-40% infill, PETG or PLA+.
- Check the slot with a scrap of gridplate before installing; the 1.03 mm top jaw is fragile.
- The exit tip thins to ~1.2 mm - handle carefully; lightly deburr so POLLEN does not catch.
- The guard is exactly 88.0 mm wide = the gap between the two upright 10-Hole U-Channels: zero clearance. Sand the edges ~0.3 mm if it binds.

STOCK vs RAIL//GUARD

What is touched - and what is not



STOCK



WITH RAIL//GUARD

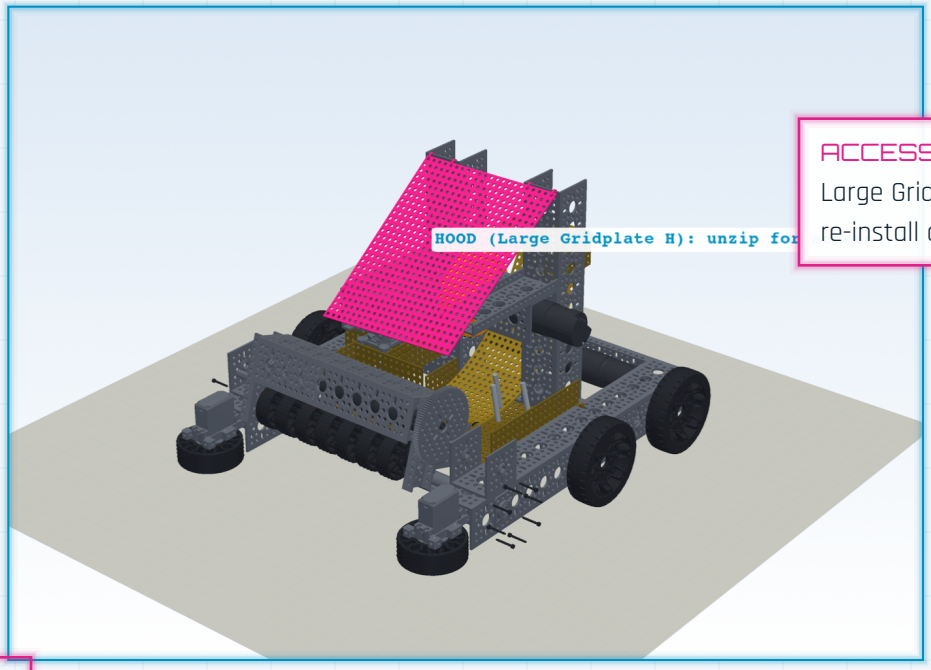
Unchanged: 1:1 Yellow Jacket, Quad Block Mount, 2-Hole U-Channel, Hyper Hub + 96 mm Hogback (wheel centre 120 mm above the axle line), drivetrain, intake, servo gate.
The guard is mounted LEVEL (flat section parallel to the ground) above the stock Medium Gridplate G (the curved shooting ramp) and becomes the new ball surface. Its flat sits 14 mm above G at the front and the curve comes down to 3-4 mm above G at the top, so G stays as the zip-tie anchor.
Note: the guard sits on the ramp under/behind the wheel, NOT where the hood (Large Gridplate H) is. The hood is only un-zipped for access and goes back on.
Removed: 1x 43 mm Round Standoff (right side, on Gridplate G) and 1 ziptie at the right front corner - both are inside the guard's footprint.

INSTALL SEQUENCE

Step-by-step

STEP 01

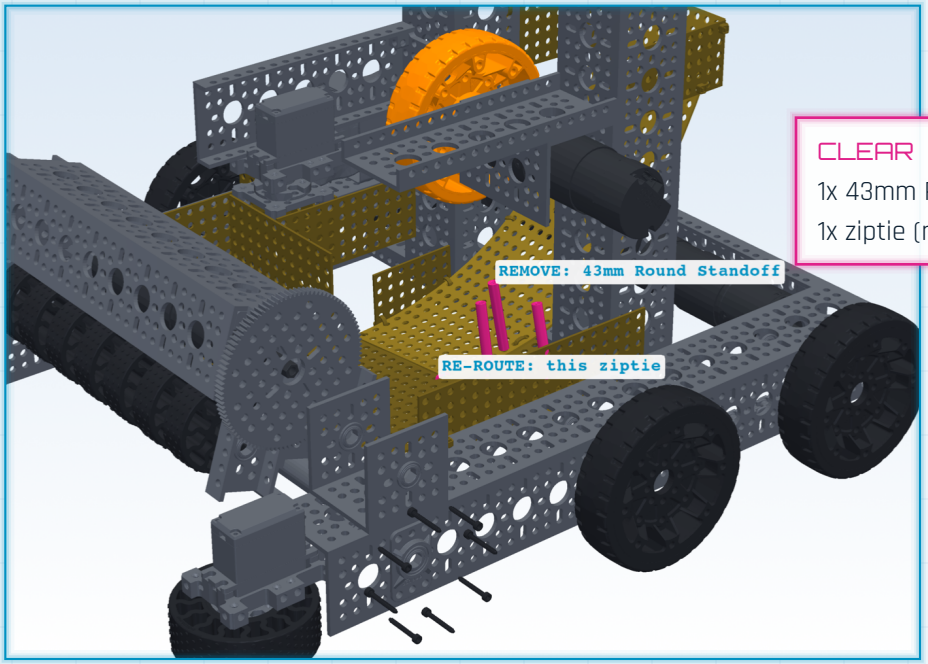
Cut the zipties on the hood (Large Gridplate H) and lift it off for access. It goes back on at the end.



ACCESS
Large Gridplate H
re-install at the end

STEP 02

Remove the 43 mm Round Standoff that stands on the right edge of Gridplate G (it passes through the guard), and cut the ziptie at the right-front corner of the ramp.



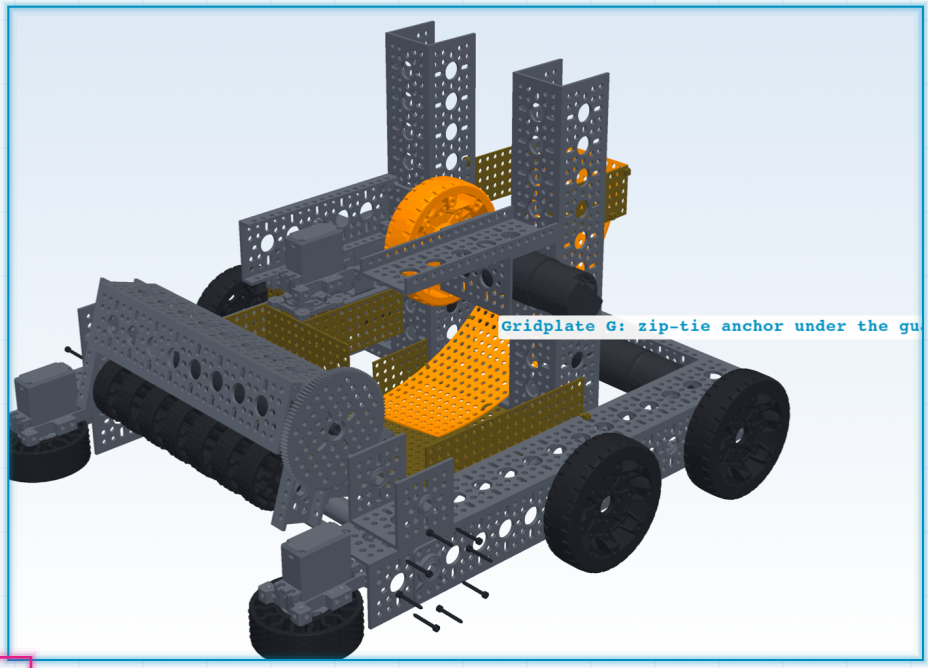
CLEAR
1x 43mm Round Standoff
1x ziptie (re-tie later)

INSTALL SEQUENCE

Step-by-step

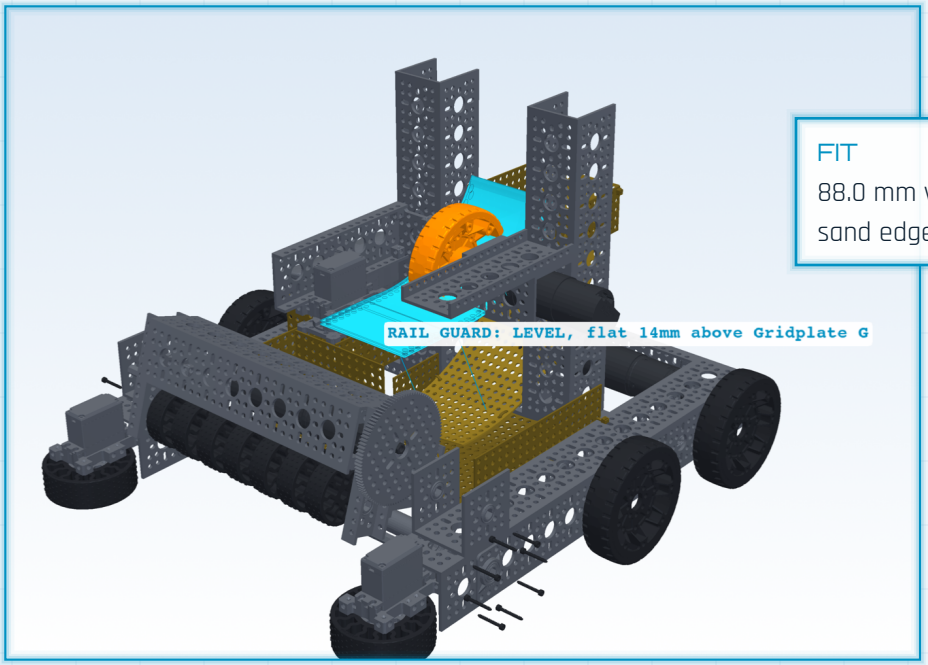
STEP 03

Gridplate G stays exactly as built. The guard does not sit on it - it is the anchor the zip ties pull down to.



STEP 04

Lower the guard between the two upright 10-Hole U-Channels, slot end at the FRONT, curve rising to the rear behind the Hogback. Keep the flat section LEVEL (check with a phone level app).



FIT

88.0 mm wide = gap between uprights

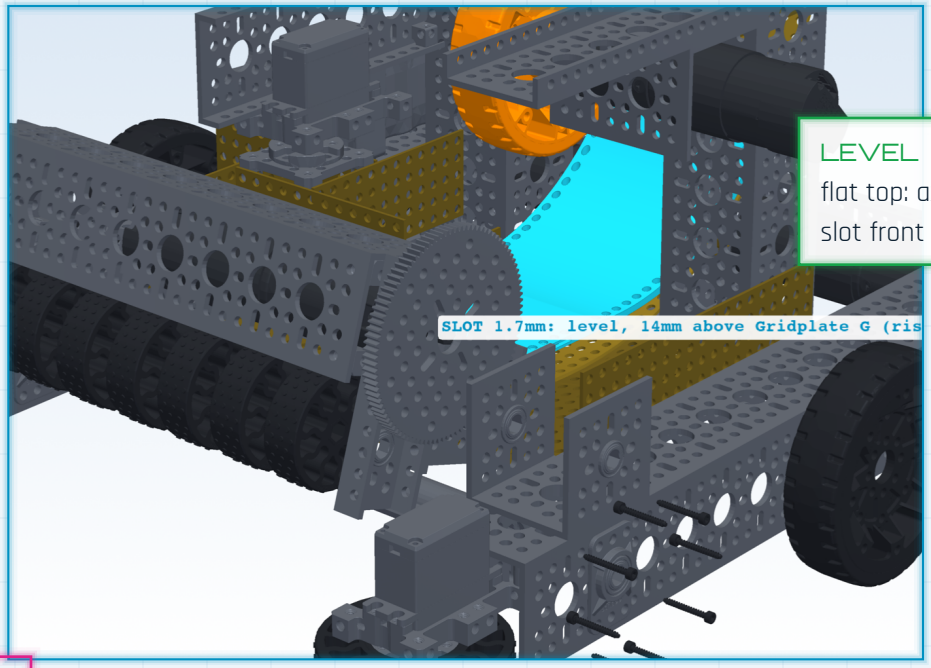
sand edges if tight

INSTALL SEQUENCE

Step-by-step

STEP 05

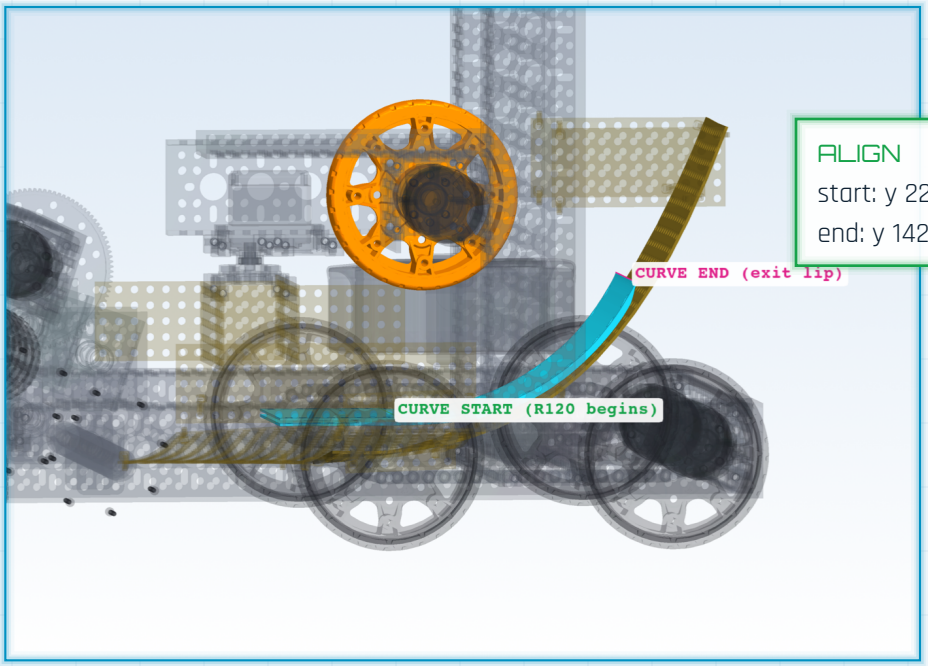
Set the height: the TOP of the flat must be 10.5 mm above the Hogback axle line (109.5 mm below the wheel centre, about 17 mm above the gridplate top). The slot then sits 14 mm above Gridplate G with no kit plate at that height: pack the gap with a riser (see open issues). Optional: one drop of superglue at the magenta spot.



LEVEL
flat top: axle line +10.5 mm
slot front 77.5 mm ahead of wheel axis

STEP 06

Front-back: the CURVE START crease (green) must be 17.5 mm behind the wheel axis; the CURVE END tip (magenta) 102.7 mm behind the axis, 78 mm above the axle line.



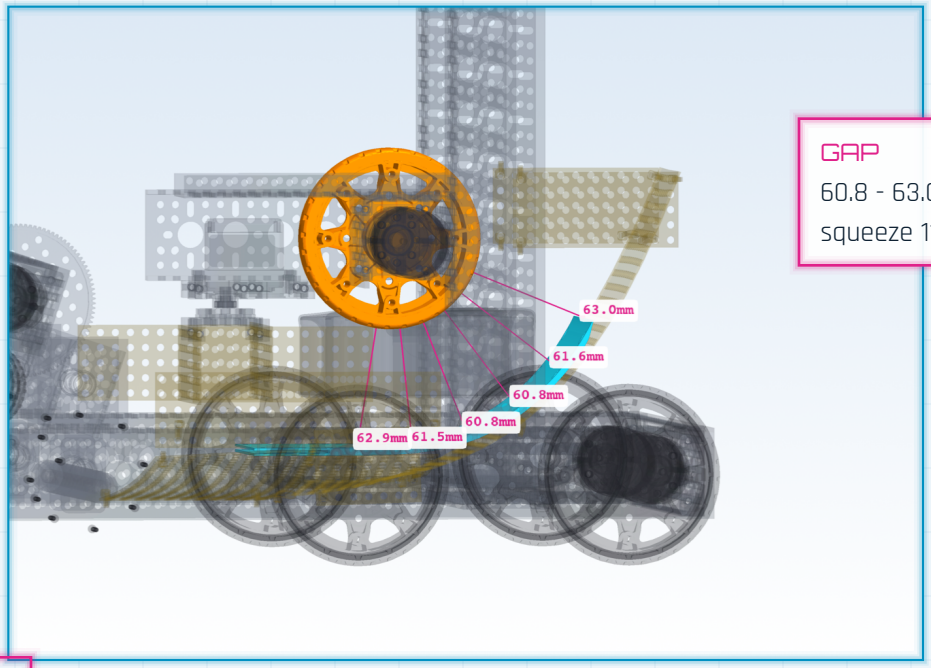
ALIGN
start: y 22.5 / z 10.5
end: y 142.7 / z 78 (kit CAD)

INSTALL SEQUENCE

Step-by-step

STEP 07

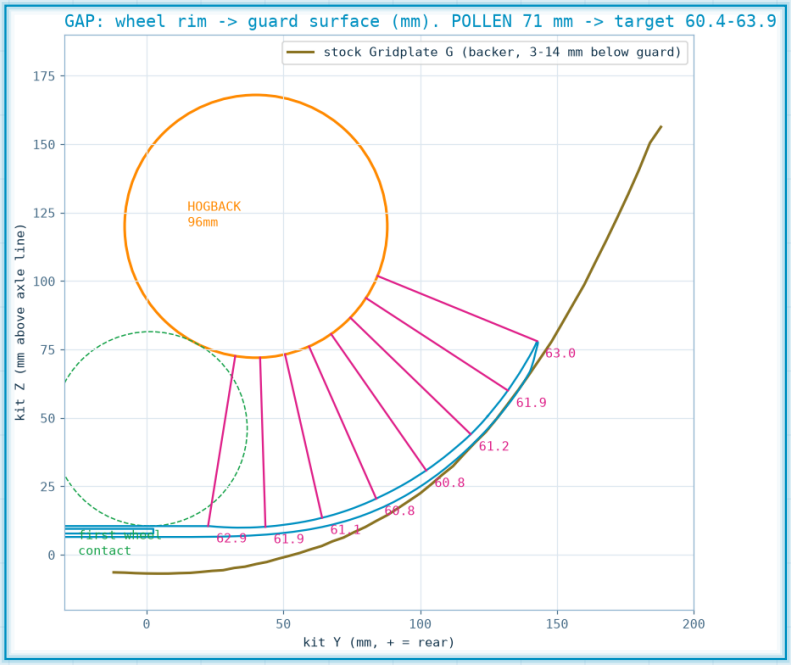
Check the gap from the Hogback rim to the guard along the arc: 60.8-63.0 mm everywhere. POLLEN is 71 mm, so it is squeezed 11.3-14.4 % and touches the whole curve.



GAP
60.8 - 63.0 mm
squeeze 11.3 - 14.4 %

STEP 08

Gap chart (side section). Blue: guard. Gold: stock Gridplate G under it. Dashed green: first wheel contact, ball still on the flat, 21 mm before the curve start.

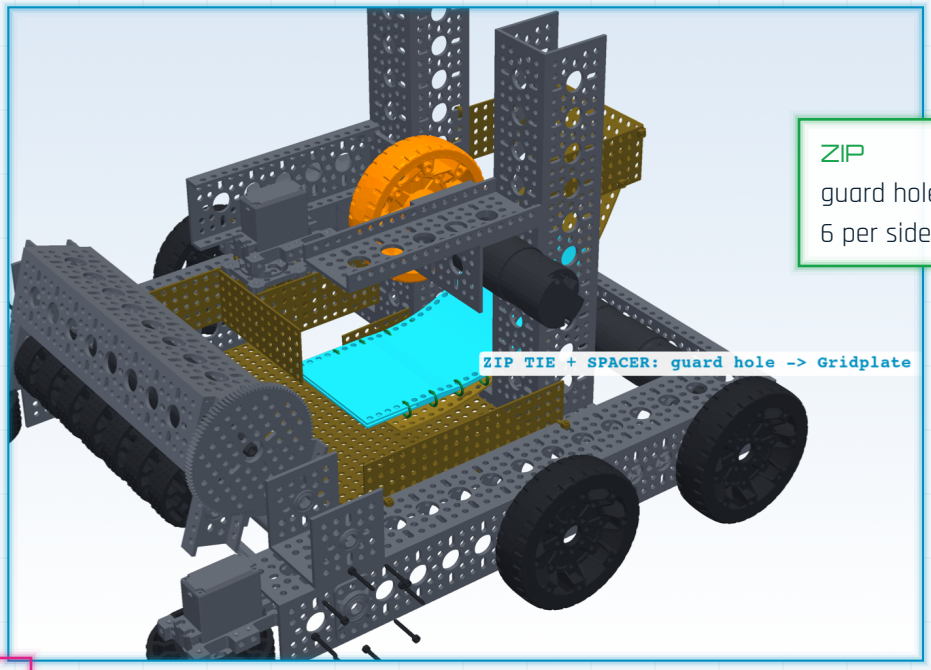


INSTALL SEQUENCE

Step-by-step

STEP 09

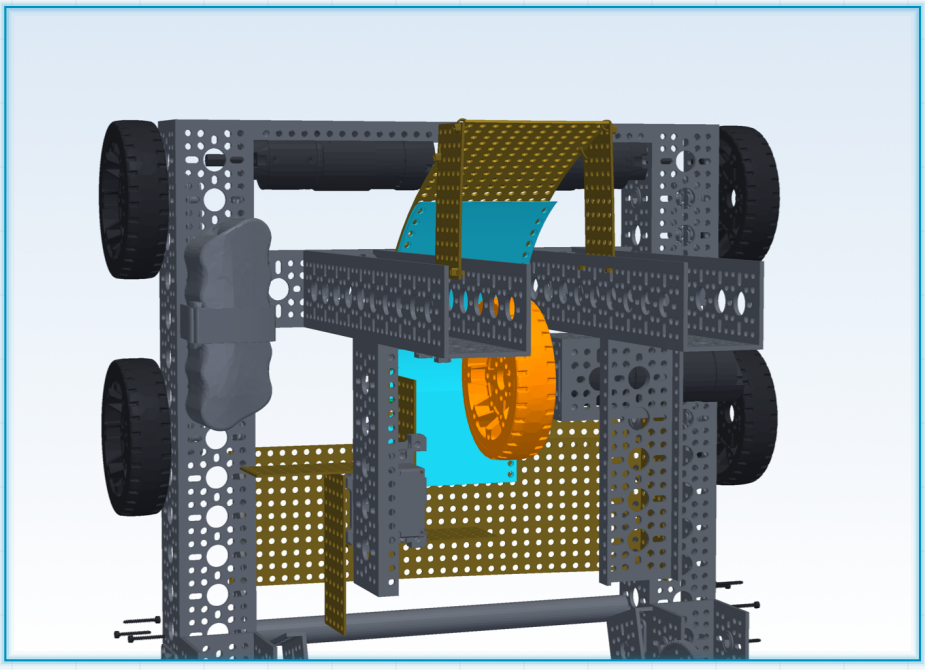
Zip-tie the guard down to Gridplate G through guard holes 4-19 (each lines up within 0-4 mm of a G hole). Put a spacer (stacked nylon washers / printed tube) on each tie so the guard is not pulled out of level: about 14 mm at the front, 3-4 mm near the top.



ZIP
guard hole -> spacer -> G hole
6 per side, snug not crushing

STEP 10

Top view. Holes 1-3 hang in front of G and holes 20-25 are 9-45 mm off any G hole - leave them empty. Only 1.9 mm of plastic around each hole: do not over-tighten. Trim tails on the underside.

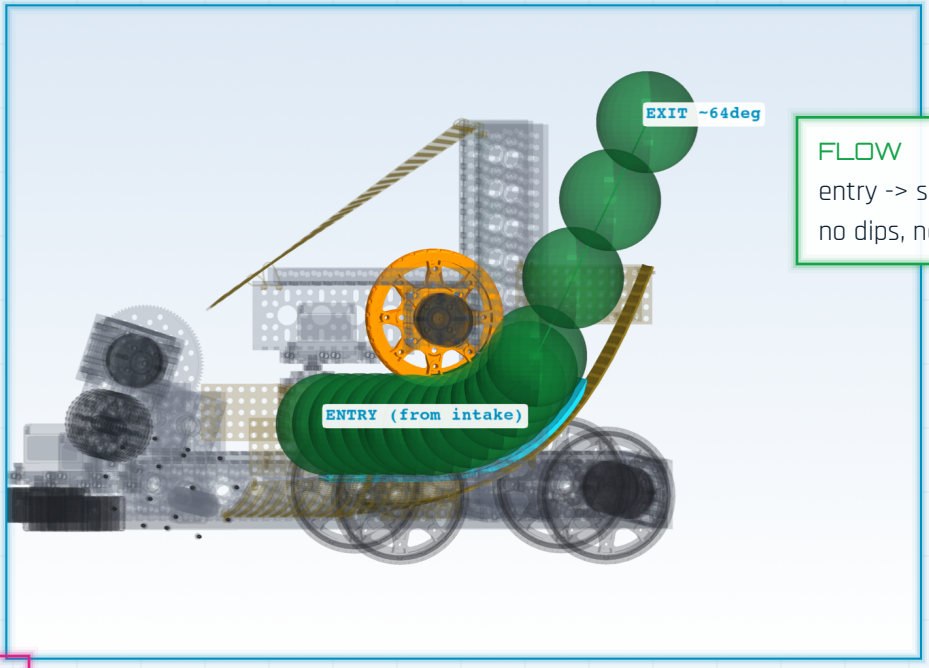


INSTALL SEQUENCE

Step-by-step

STEP 11

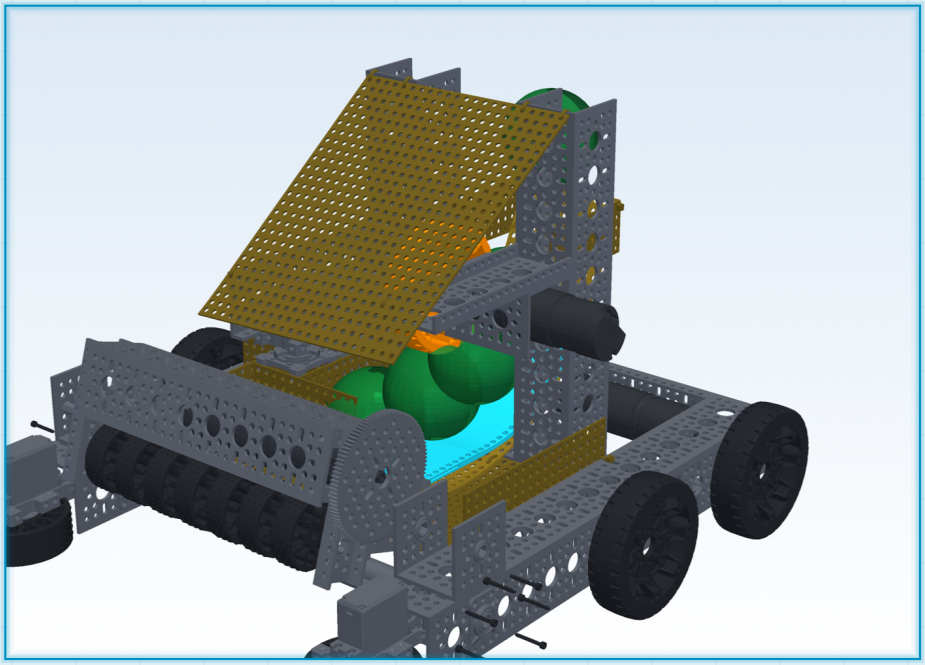
Ball path: POLLEN rolls in from the intake onto the level flat, is grabbed by the wheel, rides the 120 mm arc and leaves the tip at ~64 deg toward the rear (stock is about 60-66 deg). The exit clears Gridplate G and the hood.



FLOW
entry -> squeeze -> exit
no dips, no snag edges

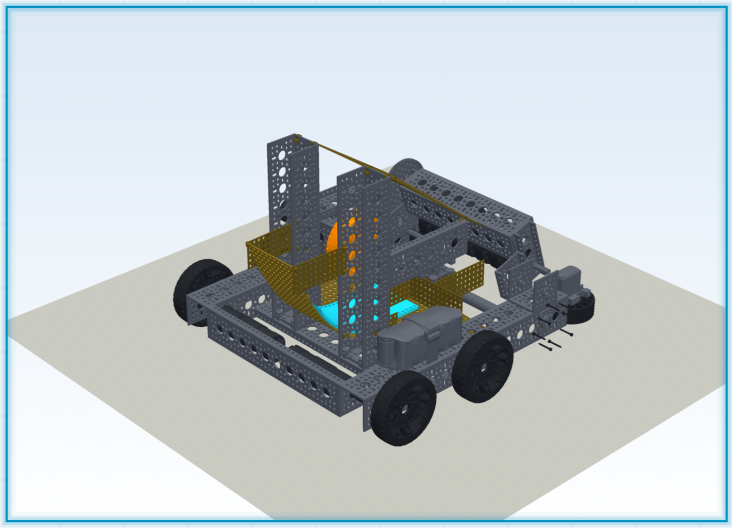
STEP 12

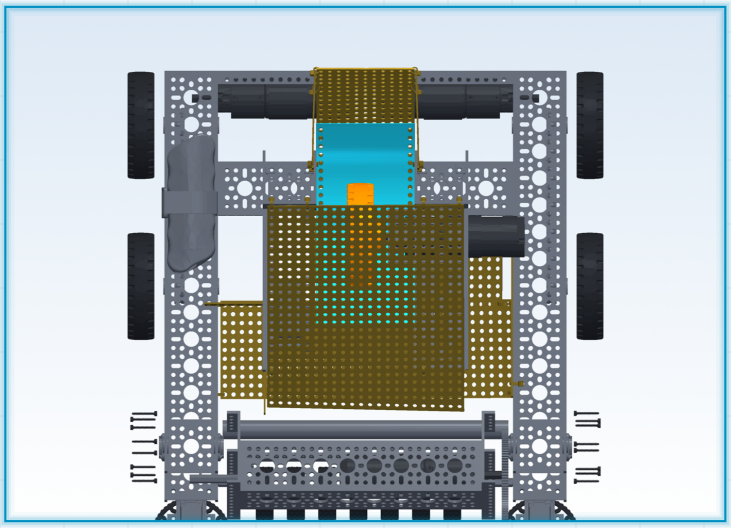
Same path in 3D. Then re-fit the hood and re-tie the right-front ziptie outside the guard.



MISSION COMPLETE

Finished robot - guard level





CAD checks (goBILDA StarterBot STEP, guard LEVEL as shown):
- Gap 60.8-63.0 mm along the whole arc = 11.3-14.4 % squeeze on 71 mm POLLEN. Closest to the Hogback: 60.7 mm.
- Clashes: the 43 mm Round Standoff (remove) and surface contact with the upright U-channels (0 mm clearance). Clear of hood, motor, hub.
- Exit ~64 deg, clears Gridplate G and the hood.

Issues found (STL not changed):
1. NO LEVEL MOUNT EDGE: at the height needed for squeeze, the slot is 14 mm above Gridplate G and 13-23 mm above the
   intake plate - nothing to clip onto. Laid level directly on G the gap is 71.5-74.7 mm = 0 % squeeze.
   STL fix: drop the flat + slot ~15.9 mm relative to the curve (a step), or add a ~14 mm riser/foot under the flat.
   Until then: zip ties + spacers hold it level.
2. No physical start/end marking lines - only a faint 5.7 deg crease.
3. 88.0 mm width = gap between the uprights (may need sanding).
4. Holes 1.9 mm from the edge; thin 1.03 mm slot jaw and ~1.2 mm tip.
5. Check with a real POLLEN and re-adjust the ties.